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# Applications, advancements, and challenges of cyanobacteria-based biofertilizers for sustainable agro and ecosystems in arid climates

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## ABSTRACT

Based on the increasing world population, the food demands have tremendously increased. Chemical-based conventional agricultural practices will be required to fulfill the food demands, which will damage the soil and create other environmental concerns. Cyanobacteria are one of the sustainable solutions to overcome agricultural struggles. With the potential to grow in extreme conditions, cyanobacteria can boost agricultural practices in arid regions by overcoming challenges such as water scarcity, elevated temperatures, salinity, and infertile soil. Cyanobacteria produce secondary metabolites that improve the soil status and bioavailability of nutrients and defend plants against biotic and abiotic stress. They are the first ones to colonize and form biological soil crust (BSC), thereby involved in ecological succession. There exists a knowledge gap related to climatic changes and their impact on terrestrial cyanobacteria as much attention is diverted towards non-terrestrial species. Covering this gap will be beneficial for the future of cyanobacteria based biofertilizer (CBF).

## 1. Introduction

With progressively increasing population, it has been anticipated by the World Health Organization (WHO) that by 2029, a 50 % increase in food production will be required to fulfill the rising food demand (Chittora et al., 2020). To achieve this, conventional chemical fertilizers will be massively used which will diminish the soil nutrients and damage the agricultural land for future use (Ghorai and Ghosh, 2022). In addition to the loss of nutritious agricultural soil, the use of chemical fertilizers also risks air contamination (Bhuyan et al., 2023). The Intergovernmental Panel on Climate Change (IPCC) states that 14 % of the total greenhouse gas is contributed solely from agriculture. Further, methane (CH<sub>4</sub>) and nitrous oxide (N<sub>2</sub>O) are approximately 47 % and 58 % respectively of which the N<sub>2</sub>O is directly associated with the use of nitrogenous fertilizers (Metz et al., 2001). Despite the use of chemical fertilizers, the productivity in arid regions is very limited due to many factors. As per the United Nations Decade for Deserts and the Fight against Desertification (2010–2020), about 41.3 % of the Earth's land is dryland which is further divided into four zones namely semi-arid (15.2

%), arid (10.6 %), sub-humid (8.7 %) and hyper-arid zone (6.6 %). These ecosystems are characterized by evaporation rates higher than the precipitation rates resulting in water shortage which further leads to drought and enhanced climatic variation (Trenberth, 2011). Accordingly, these factors are limiting conventional agriculture, hence novel strategies are required to overcome those challenges and assist in feeding the increasing worldwide population.

The Green Revolution aims to decrease the use of chemical fertilizers and instead relies on biofertilizers (John and Babu, 2021). Considering the adverse effects of conventional fertilizers on both the environment and human health, it is preferred to use biofertilizers to prevent any health and environmental hazards. In the Green revolution/technology, the microbes are exploited indirectly to produce food by improving crop quality as well as its yield (Pathak et al., 2017). It has become a promising approach for improving soil and plant health through escalating nutrients, reducing dependency on artificial fertilizers, and mitigating handling-based plant diseases (Múnera-Porras et al., 2020). Fig. 1 shows a comparison between conventional agricultural practice and the green revolution.

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Cyanobacteria are prokaryotic photoautotrophic microbes, however, some are facultative heterotrophs with the ability to utilize different carbon (C) and organic nitrogen (N) sources (Pham et al., 2017). Cyanobacteria have many benefits due to which it can be used as a potential biofertilizer. They can increase soil porosity, decrease soil salinity, and produce adhesive substances such as exopolysaccharides (EPS) (Thapa et al., 2023). They produce different secondary bioactive molecules, amino acids, and vitamins as well as phytohormones such as auxins (Kollmen and Strieth, 2022). Moreover, it improves the water retention capacity of soil. Cyanobacteria are also involved in bioremediation through different mechanisms thereby, playing a dual role (Chittora et al., 2020). This review will focus on the challenges that limit agricultural activities in arid and semi-arid regions. We will describe the advantages of using CBF to improve plant productivity through its biocontrol activities and improvement of soil nutritional value by enriching soil elements and remediation of pollutants. This review will also highlight the importance of cyanobacteria in ecological succession and restoration and would also elaborate on the cyanobacterial responses to climatic changes and the implications in arid environment. Lastly, the review will discuss the limitations that decelerate or restrict the use of cyanobacteria.

## 2. Challenges limiting the agriculture sustainability in the arid and semi-arid regions

### 2.1. Freshwater scarcity

Water scarcity stands as the major challenge directly hindering agricultural development in arid regions. The agricultural practices of people living in arid and semi-arid regions depend on the available natural resources. In such regions, the precipitation is low due to which droughts are persistent (Mavhura et al., 2015). Regions where the annual precipitation is below the minimum crop water requirement, there is a possibility that crops would undergo water stress at a phase during its growth and developmental cycle (Kattumuri et al., 2017). Under drought stress, crops yield decreases, for instance, soybean and

maize yield is at risk of 70 % decrease in its yield whereas wheat and rice have a yield decrease risk of about 68 % and 64 % correspondingly (Leng and Hall, 2019). Not only drought but flooding can also occur in such dry regions due to poor vegetation cover, therefore, intensified precipitation even for short period results in flooding (Liu et al., 2022; Shen et al., 2020). Drought exerts a significant impact on nutrient levels and primary production, leading to soil predominantly becoming alkaline, characterized by elevated pH levels and high concentrations of potassium (K), calcium (Ca), salts, and other minerals that could directly influence plant growth and development (Bista et al., 2018; Jiao et al., 2016).

### 2.2. Temperature

The arid and semi-arid regions are marked by hot climatic conditions with intense solar radiation (Mussard and Amara, 2018), hence high mean temperatures during the daytime. With an increase in temperature, the evapotranspiration also increases due to which more water is required to maintain the health and productivity of crops (Alori et al., 2020). Under elevated temperature and light intensity, plants might undergo photoinhibition, which is the absorption of light above the required threshold leading to reduced photosynthetic rate (Jacobsen et al., 2012). The nutrient acquisition and photosynthetic productivity of plants decline when they undergo temperature stress. This type of stress affects the plant developmental stages and associated growth period such as grain filling and tillering (Hossain et al., 2013). Moreover, the metabolic process of plants under heat stress is disturbed thereby reducing the life cycle of plants (Kumar and Verma, 2012). Extreme temperatures in the arid regions effects the nutrient cycling processes (Hag Husein et al., 2021).

### 2.3. Salinity

Soil salinity is one of the major limitations in arid and semi-arid regions and impacts plant growth and development negatively. In particular, it has been reported that salinization has adverse effects on

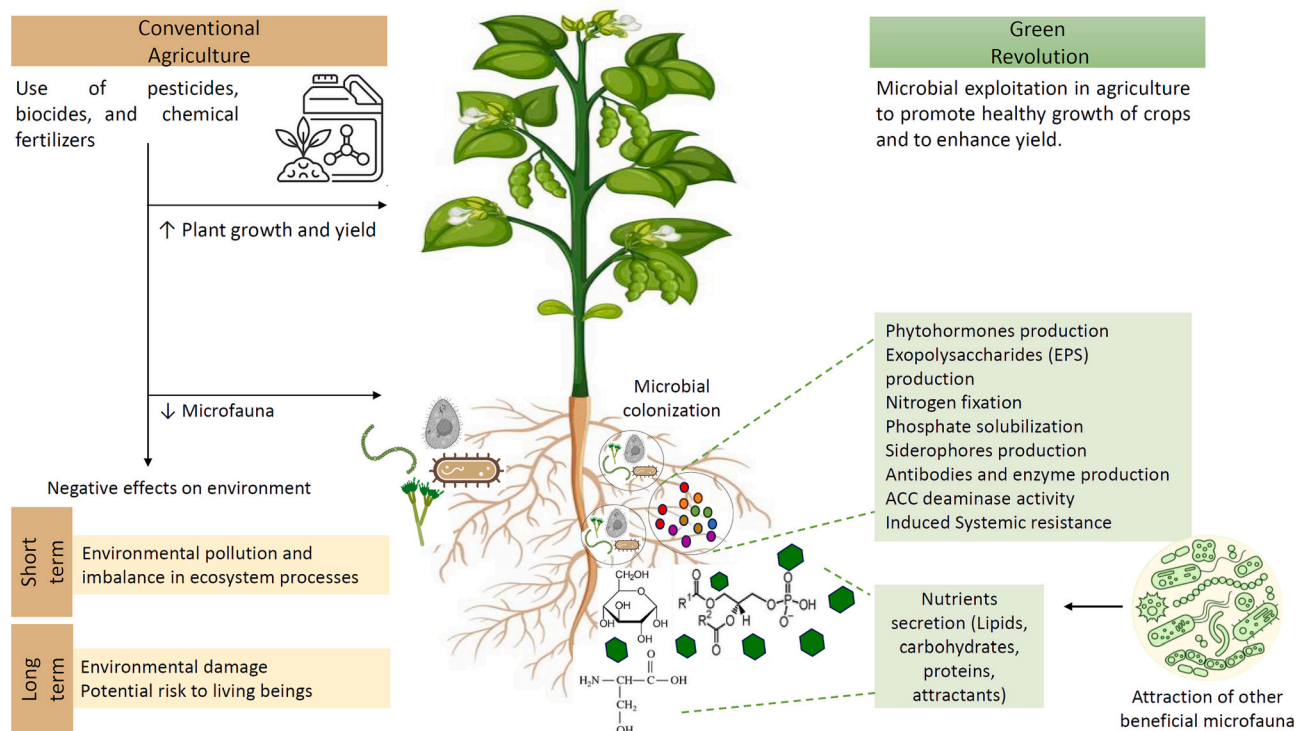


Fig. 1. Comparison between conventional agriculture and green revolution.

the agricultural industry, resulting in a net annual loss of approximately 27.3 million dollars (Wichelns and Qadir, 2015). When soil salinity levels are high, the concentration of sodium ( $\text{Na}^+$ ) and chloride ( $\text{Cl}^-$ ) ions exceed the maximum possible levels, disrupting not only the osmotic functions of plants but also the metabolic processes necessary for their growth (Balasubramaniam et al., 2023). Salinity stress does not solely entail ionic imbalance, it also affects the photosynthetic and respiratory processes by generating reactive oxygen species (ROS) (Jacoby et al., 2010; Miller et al., 2010).

Furthermore, both salinity and drought conditions can lead to lower N concentrations, affecting N uptake by plants (Wang et al., 2023). Conversely, phosphorus (P) is often found in higher concentrations in soil, yet its bioavailability remains limited (Fageria et al., 2011). Also, it is worth noting that agricultural and natural soils found in arid regions generally exhibit concentrations of metallic micronutrients such as iron (Fe) and zinc (Zn). With these micronutrient levels being low or almost inadequate, the expected ecosystem services from arid regions are on the decline.

#### 2.4. Soil properties

Soil properties will be described in the context of soil carbon pool and the soil nutrients status.

##### 2.4.1. Soil carbon pool

The level of soil organic carbon (SOC) and soil inorganic carbon (SIC) is considered as one major issue related to arid environments. SIC plays a role in aggregating soil particles by interacting with soil organic matter (SOM) (Naorem et al., 2023). Soil texture and structure plays a major role in determining the SOC as its aggregation helps in C capture and storage (Merante et al., 2017). The soil in arid regions is sandy characterized by high water permeability, low water holding capacity, low nutrient holding capacity, limited nutrient exchange capacity, poor soil structure, and high vulnerability to wind erosion (Osman, 2018; Wambeke, 1992). Conventional agricultural practices rely on the use of synthetic fertilizers, pesticides, extensive tillage, and irrigation. These practices have helped improve crop productivity while simultaneously deteriorated the soil fertility (Francaviglia et al., 2023).

##### 2.4.2. Low nutrient levels

The soil of arid regions has low N, C, and P content along with low availability of Zn, Fe, and molybdenum (Mo) due to the formation of insoluble calcium phosphates (apatite) (Moreno-Jiménez et al., 2023). The presence of these insoluble carbonates interferes with the chemistry and availability of both macro and micro-nutrients by increasing the soil alkalinity (Brady et al., 2008). The challenges limiting agriculture

sustainability are presented in Fig. 2.

Sustainable agriculture aims to exploit the potential of natural resources and microbiota to produce biofertilizers that can address these challenges. Several microorganisms, such as bacteria, fungi, and algae, have been used and commercialized to produce biofertilizers to enhance crop production and soil quality while also helping to alleviate biotic and abiotic stressors (Chaudhary et al., 2022). Among them, cyanobacteria represent the best candidate due to several advantages that will be discussed in this review article.

### 3. The beneficial role of CBF for plants and soil

Cyanobacterial biomass is widely used for the production of bio-fuels, superfoods (food supplements), and bio-fertilizers (Benson et al., 2014) due to its reported harmlessness and eco-friendly characteristics. Advantages of cyanobacteria encompass nitrogen fixation, organic waste digestion, detoxification of pollutants, and production of bioactive compounds (Ammar et al., 2022; Żyłańczyk-Duda et al., 2022). These microorganisms contribute to soil enrichment, positively influencing plant growth by enhancing nutrient availability, forming complexes with pollutants, and facilitating organic molecule mineralization (Osorio-Reyes et al., 2023; Shukia et al., 2008). Additionally, cyanobacteria help maintain soil moisture, sustain soil pH through nitrogen fixation, and promote soil structure improvement by producing EPS and plant growth promoting substances (PGPS) (Singh et al., 2016). Loose to tight associations with plant roots are formed based on cyanobacteria invasiveness (Karthikeyan et al., 2007). Some of the benefits of using CBF are illustrated in Fig. 3.

#### 3.1. Direct stimulation of the plant growth and productivity

Cyanobacteria-based biofertilizers improve plant growth directly by enhancing the soil nutrients mainly N. This is achieved through the formation of a symbiotic association between plants and cyanobacteria known as the cyanobiont. This section will discuss how cyanobacteria forms a healthy association with plants and undergoes several morphological and functional changes to improve plants' productivity. The use of biofertilizers is beneficial in various ways. Most significant one being is the reduction in the amounts of chemical fertilizers used. Biofertilizers are natural sources of N through the nitrogen-fixing capabilities of microbes, such as cyanobacteria. Many important species of cyanobacteria such as *Nostoc* and *Anabaena* are known to fix higher concentrations of  $\text{N}_2$  (Kishore Choudhary and Bimal, 2010). The fixed  $\text{N}_2$  is then available in the surrounding environment in different forms such as vitamins, ammonia, and amino acids. Cyanobacteria can fix approximately 22.3 to 53.1 kg  $\text{Nha}^{-1}$  which could reduce the use of chemical

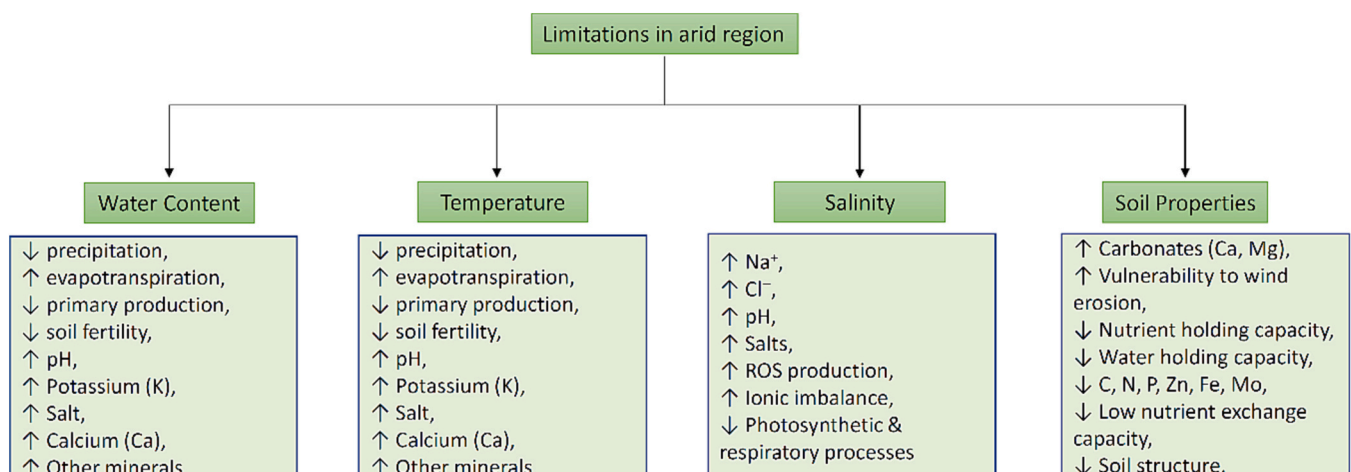


Fig. 2. Challenges limiting the sustainability of conventional agriculture.



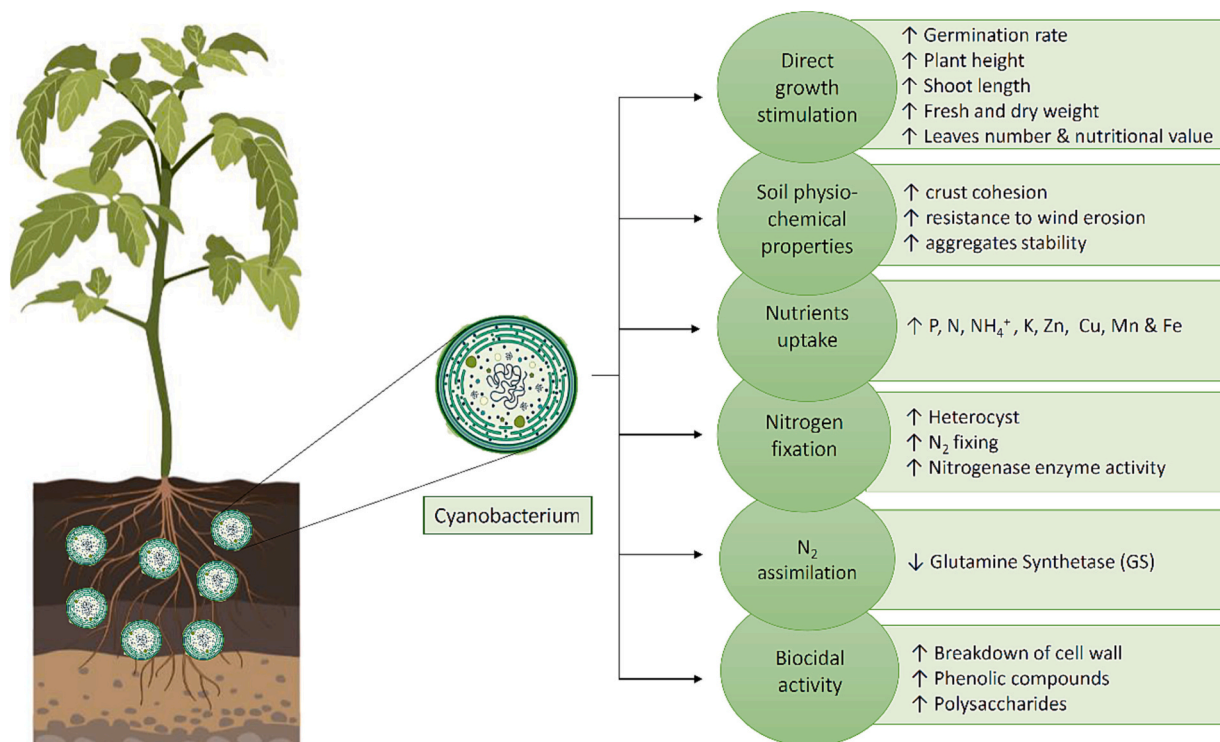


Fig. 3. Potential benefits of using cyanobacteria-based fertilizer in agriculture.

fertilizer by 25 % to 50 % (Issa et al., 2014). Another study reports that cyanobacteria can fix 10 to 30 Kg  $\text{Nha}^{-1}$  per year (Ghorai and Ghosh, 2022).

Cyanobacteria as biofertilizers to promote crop production has been extensively studied. *Arthrospira platensis* is known to improve the yield of rice by 20.9 % and enhance growth parameters such as height, total leaf area, fresh and dry biomass (Dineshkumar et al., 2018). Similarly, another research has demonstrated positive results in terms of plant growth, with an approximate 28 % increase in height (Khan, 2018). The role of cyanobacterial species *Anabaena laxa* and *Calothrix elenkinii* were analyzed on coriander, cumin, and fennel seeds. The results showed that *Anabaena* enhanced the seed germination of cumin as compared to the control while *Calothrix* improved the length of roots and shoots of all three crops, more specifically of fennel seeds. Moreover, in coriander,  $\beta$ -1, 3 endoglucanase activity improved by 2-fold in shoots when inoculated with *Anabaena* (Kumar et al., 2013). This protein in plants is mainly involved in fighting against fungus. Similarly, in another research, *Anabaena laxa* is used to enhance the growth and yield of *Cicer arietinum* L. (Chickpea). With the inoculation of the cyanobacterium, the leghemoglobin content and the biomass of chickpea were enhanced by 2-folds (Bidyarani et al., 2016). Leghemoglobin is a protein used to carry oxygen ( $\text{O}_2$ ) in the nodules of the roots of  $\text{N}_2$ -fixing plants. Moreover, there was a positive correlation reported between the nutrient uptake by the plant and its improved growth and development with the cyanobacterium activities such as  $\text{N}_2$  fixing, enhanced chlorophyll content, and bioavailability of N in the soil. The inoculated chickpea accounted for an improved yield of grains by 50 % (847  $\text{kg ha}^{-1}$  to 1724  $\text{kg ha}^{-1}$ ) (Bidyarani et al., 2016).

N is a major limiting and costly factor in agriculture (Anas et al., 2020; Rütting et al., 2018). To address the need for N in the soil, fertilizers are typically recommended; however, approximately half of these fertilizers are not absorbed or utilized by crops and end up being wasted within the environment (FAOSTAT, 2020). Biofertilizers based on nitrogen-fixing cyanobacteria (NFC) are also widely used due to their efficiency in improving the net crop yield. For instance, NFC used to inoculate rice fields improved the yield by 25 % to 34.6 % (Dash et al.,

2016). Another positive aspect of adding NFC is to reduce the addition of chemical N sources which reportedly saved an average of 45  $\text{kg ha}^{-1}$  (Kollah et al., 2016; Prasanna et al., 2015a). NFC is not only helpful in improving the crop's health but is also known to enhance the status of flooded rice fields and crops in paddy soil (Dron et al., 2012; Hashtroudi et al., 2013; Prasanna et al., 2015a). SOC and soil total nitrogen (STN) are also improved by the inoculation of NFC which ultimately enriches the soil community structure. Improved SOC helps in the stabilization of soil by developing useful physiochemical properties thus, promoting nutrient cycling (Leong et al., 2021; Wang et al., 2015).

The studies references suggest that employing cyanobacteria, particularly the N fixers can boost crop yield and decrease expenses related to N inputs.

Table 1 below reports some of the recent studies that utilized cyanobacteria along with other microbes to improve the growth of onion, wheat, cowpea, licorice, tomato, and basil by enhancing certain growth factors. Recent research findings elucidate that cyanobacteria have the potential to emerge as a sustainable and green method in the agricultural sector soon.

### 3.2. Prime features of symbiotic association of cyanobacteria with plants.

All naturally occurring cyanobacteria cannot form symbiotic associations. Some can colonize the roots and shoots of plants and carry out symbiosis. Moreover, the most important enzyme; nitrogenase is reported to have the maximum efficiency when in colonized state as compared to the free-living state of the cyanobacteria. This symbiotic relationship, once established, stays stable under different environmental conditions as a complex signaling and sensing system is activated to maintain a synchronized association in terms of the growth, and development of the respective parties. The process begins with a controlled infection of the host by the cyanobacterium followed by the transport of nutrients and support in cell division, in return, the plant host maintains the cyanobacterial biomass (Tikhonovich and Provorov, 2007). The prime function of this symbiotic association for plants is the uptake of fixed  $\text{N}_2$  through the symbiont while for the symbiont, it is

**Table 1**

Recent studies using cyanobacteria for the improvement of plants growth promoting factors.

| Cyanobacteria                                                                                 | Plant                                    | Effect                                                                                                                                                                                                                                                                                                                                                        | Reference                           |
|-----------------------------------------------------------------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <i>Arthrospira platensis</i> & <i>Lithothamnium</i>                                           | Onion                                    | ↑ 52 % root fresh mass,<br>↑ 124 % root dry mass,<br>↑ ~34 % leaves sugar level,<br>↑ 21 % total free amino acids,<br>↑ ~52 % Nitrate reductase (NR) activity                                                                                                                                                                                                 | (de Oliveira Amatussi et al., 2023) |
| <i>Leptolyngbya</i> sp. RBD05 & <i>Paenibacillus sabinae</i>                                  | Wheat                                    | ↑ 26 % K content,<br>↑ 85 % dry weight,<br>↑ 70 % root length                                                                                                                                                                                                                                                                                                 | (Duan et al., 2023)                 |
| <i>Nostoc mucorum</i> , <i>Bacillus amyloliquefaciens</i> , & <i>Saccharomyces cerevisiae</i> | Cowpea                                   | ↑ 390 % dehydrogenase enzyme activity,<br>↑ 180 % chlorophyll levels,<br>↑ 130 % dry weight,<br>↑ 68 % pod length,<br>↑ 190 % pod dry weight,<br>↑ 16.6 % midrib zone thickness,<br>↑ 22.6 % vascular bundle length,<br>↑ 42.4 % upper epidermal leaf layer,<br>↑ 33.5 % lower epidermal leaf layer,<br>↑ 36.9 % palisade tissues,<br>↑ 26.5 % spongy tissues | (Omer et al., 2023)                 |
| <i>Anabaena sphaerica</i> & <i>Nostoc pruniforme</i>                                          | Licorice ( <i>Glycyrrhiza glabra</i> )   | ↑ 100 % seed germination rate,<br>↓ 60 % in hydrogen peroxide (H <sub>2</sub> O <sub>2</sub> ) accumulation,<br>↑ 25 % catalase (CAT) activity,<br>↑ 13 % peroxidase (POX) activity,<br>↑ 65 % accumulation of glycyrrhizin acid (GA),<br>↑ 70 % accumulation of licoirigenin (LG)                                                                            | (Abbasi and Mohammadi, 2023)        |
| <i>Arthrospira platensis</i>                                                                  | Tomato ( <i>Solanum lycopersicum</i> L.) | ↑ 87 % roots length,<br>↑ 757 % roots weight,<br>↑ 47 % number of leaves,<br>↑ 137 % size of leaves,<br>↑ 121 % chlorophyll a,<br>↑ 96 % chlorophyll b,<br>↑ 130 % carotenoids                                                                                                                                                                                | (Fadia et al., 2023)                |
| <i>Wolleea vaginicola</i> ISB89 and <i>Nostoc spongiaeforme</i> var. <i>tenuis</i> ISB91      | <i>Mentha pulegium</i>                   | ↑ 24 % and 96 % root length,<br>↑ 23 % and 92 % shoot length,                                                                                                                                                                                                                                                                                                 | (Riahi et al., 2023)                |

**Table 1 (continued)**

| Cyanobacteria                                                                                                                         | Plant                                    | Effect                                                                                                                                                                                                                                                             | Reference                  |
|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
|                                                                                                                                       |                                          | ↑ 107 % and 128 % fresh weight of aerial parts,<br>↑ 93 % and 77 % dry weight of aerial parts,<br>↑ 233 % and 417 % fresh root weight,<br>↑ 58 % dry root weight,<br>↑ 229 % and 129 % stem numbers,<br>↑ 45 % number of leaves                                    |                            |
| <i>Leptolyngbya</i> sp.                                                                                                               | Basil ( <i>Ocimum basilicum</i> L.)      | ↑ 34.4 % plant growth,<br>↑ 30 % shoot fresh weight,<br>↑ 36 % root fresh weight,<br>↑ 24 % number of leaves                                                                                                                                                       | (Santini et al., 2022)     |
| <i>Roholtiella</i> sp.                                                                                                                | Bell pepper ( <i>Capsicum annuum</i> L.) | ↑ 17.014 % shoot length,<br>↑ 39.15 % fresh weight,<br>↑ 31.02 % dry weight,<br>↑ 92.05 % Relative Water Content (RWC)                                                                                                                                             | (Bello et al., 2022)       |
| Combined extracts ( <i>Dunaliella salina</i> , <i>Chlorella ellipsoidea</i> , <i>Aphanothece</i> sp., and <i>Arthrospira maxima</i> ) | Tomato                                   | ↑ 22 % shoot length,<br>↑ 87.50 % root weight,<br>↑ 70.19 % shoot weight,<br>↑ 16.88 % chlorophyll a,<br>↑ 13.55 % chlorophyll b,<br>↑ 16.62 % β-carotene,<br>↑ 24.19 % SOD,<br>↑ 97.59 % POD,<br>↑ 182.95 % N uptake,<br>↑ 78.35 P uptake<br>↑ 165.13 % K uptake, | (Mutale-joan et al., 2021) |

protection from wide environmental variations along with constant nutrient provision.

The cyanobiont undergoes several structural (morphological) and functional changes (growth, photosynthetic) to fulfill the needs of the host. The morphology of heterocyst is naturally designed in a way that allows efficient working of nitrogenase enzyme to fix N<sub>2</sub>. The structure protects it from O<sub>2</sub> which is toxic for the enzyme nitrogenase (Issa et al., 2014). Functionally, the frequency of heterocyst increases which is the N<sub>2</sub> fixation site. The mechanism of heterocyst increase in both, free-living cyanobacteria and cyanobiont is different (Álvarez et al., 2023). In the former, heterocyst frequency increases when there is N<sub>2</sub> starvation while in the latter, the frequency increases when no N<sub>2</sub> starvation is detected during symbiosis. Moreover, the differentiation of heterocyst also depends on the C concentration of cyanobacteria. If the cyanobiont provides both C and N to the host, the heterocyst differentiation and overall increase in frequency does not occur (Rai et al., 2019). Hence, it is valid to state that the whole status of the cyanobiont plays a major role in driving the mechanisms. Further the expression of the gene responsible for nitrogenase enzyme increases which enhances N<sub>2</sub> fixation. On

the contrary, the ammonia assimilating glutamine synthetase (GS) enzyme activity decreases. The concentration and activity of GS inside the heterocyst is more than the vegetative cell in cyanobacteria. For the assimilation of N from ammonia, the gene expression of GS is required to be higher. The highest GS activity is noticed in angiosperms followed by bryophytes, then genus *Azolla*, and lastly lichens. The decrease in GS activity decreases the N<sub>2</sub> assimilation, and hence, liberates the N<sub>2</sub> to the hosts. All the transfers of C and N occur at a specific cyanobacteria-host interface, a point where cyanobacteria are in close contact with the host (Adams et al., 2013).

Certain cyanobacterial species such as *Nostoc*, and *Anabaena* are studied for their efficient capabilities of improving the fertility of the soil. Some studies have utilized them for bioremediation purposes (Li et al., 2019). In the study carried out by Ranjan and colleagues (2016), cyanobacterial inoculation was reported to enhance nitrogenase activity by 48 % and plant biomass by 70 % (Ranjan et al., 2016). In other words, cyanobacteria can be helpful in multiple ways such as improving soil fertility and plant minerals uptake, production of plant growth hormones, and removal of pollutants from soil.

### 3.3. The role of CBF on plant tolerance to abiotic stresses

Inoculation of cyanobacteria in soil improves its fertility by two mechanisms (Chamizo et al., 2018) (i) releasing extracellular substances that include peptides, polysaccharides, and lipids that are distributed around the soil particles. These substances hold the soil particles together forming stable aggregates; (ii) the fiber within polysaccharides expands and entangles the soil clay particles forming macroaggregates. These macroaggregates join to form bigger soil aggregates. These interconnected aggregates formed by the provision of cyanobacteria not only improve the soil stability but also enhance the organic C content within the soil (Kheirfam et al., 2017). Furthermore, cyanobacteria reduce wind erosion and enhance the water-holding capacity of soil. Cyanobacteria are one of the best candidates to be used for improving the soil status in regions with arid climates. This could be attributed to the versatile nature of cyanobacteria by their ability to grow and survive in different environmental conditions with tolerance to extreme conditions like drought, salinity, and high temperature (Pathak et al., 2018).

#### 3.3.1. Role of secondary metabolites and phytohormones in absorbing light

The secondary metabolites produced by cyanobacteria make them potential candidates to be used as biofertilizers. Cyanobacteria are rich in pigments like chlorophyll *a*, carotenoids, and phycobiliproteins. The phycobiliproteins include phycoerythrin (PE, pink pigment), phycocyanin (PC, blue pigment), and allophycocyanin (APC, blue-grey pigment) and makeup 20 % of the dry weight of cyanobacteria (Haddad et al., 2020). These pigments absorb light of different wavelengths and have complex structures. These pigments have a backbone made up of isoprene and tetrapyrrole rings, however, carotenoids are composed of isoprene units only. Among the listed pigments, chlorophyll is the most prominent light-harvesting pigment which has six distinct types that are: Chlorophyll *a*, *b*, *d*, *f*, and divinyl chlorophyll *a* (chl *a*2) and *b* (chl *b*2) (Saini et al., 2018). Chlorophyll is one of the most significant photoreceptors owing to the single and double bonds within its structure that allow delocalization of electrons. This feature of delocalization within the structure enables the chlorophyll to absorb light from different bands of the visible spectrum of light and initiate the transport of electrons to carbon dioxide (CO<sub>2</sub>) via the electron transport chain (ETC) (Song et al., 2015). Carotenoid is another light-harvesting pigment found along with chlorophylls and functions in absorbing light from the visible spectrum which the chlorophylls could not absorb (Cerezo et al., 2012). As described earlier, the structure of the carotenoid is mainly composed of isoprene units. Carotenoids can be divided into different groups based on the presence or absence of an oxygen atom (O) at the terminal end of the structure. If the O is present, the derivatives are known as xanthophylls while the deoxygenated derivatives are known as

carotene (Maoka, 2020). Carotenoids are commercially important because of their antioxidant properties, colors, and health-related factors (Elvira-Torales et al., 2019). Phycobiliproteins are other pigments found in cyanobacteria that are colored and are tetrapyrrole biliprotein. These pigments absorb light from the visible spectrum where the absorption capacity of chlorophyll is the least. Phycobiliproteins are organized in a specific arrangement and are further grouped based on the absorption spectrum. PE, PC, and APC have absorption at 565, 620, and 650 nm (Kini et al., 2020; Saini et al., 2018) respectively.

Phytohormones produced by cyanobacteria play a crucial role in crop yield by improving plant tolerance to different environmental stressors. Some important phytohormones are gibberellins, auxins, and cytokinin (Gonçalves, 2021). Indole 3 Acetic Acid (IAA), a secondary metabolite from the indole alkaloids, is a very well-known auxin for its role in growth promotion and in fighting against plant pathogens (Chieb and Gachomo, 2023). IAA is an important auxin as it is involved in different regulatory and growth pathways in plants. Cyanobacteria produce IAA and promote plant growth when they are in close association. In literature, most of the research focuses on the use of bacteria for plant growth promotion while a very small portion of this field is attributed to cyanobacteria despite its ability to produce secondary metabolites or efficient bioactive compounds (Ahmed et al., 2010). Siderophores are another important secondary metabolite with a high affinity for Fe. Siderophores improve the bioavailability of Fe for plants uptake (Albelda-Berenguer et al., 2019). These secondary metabolites are organic in nature and are very helpful in chelating the ferric form of Fe, especially under conditions with extremely low Fe levels. In other words, siderophores or siderophore-releasing microorganisms help in enhancing the bioavailability of Fe so that it can easily be absorbed by other biological entities, especially plants (Toribio et al., 2020). Amino acids or polyamines produced by cyanobacteria are known to diminish the injuries caused by environmental stressors in the host plant. An example of the amino acid involved in plant defense and growth is L-amino acids. These amino acids act as signaling molecules in defense mechanisms, improvement of sugar concentration, and have role in the synthesis of chlorophyll (Santini et al., 2021). A variety of polysaccharides are found in the cyanobacteria. A study conducted in Morocco applied polysaccharides extracted from cyanobacteria on plants. The study reported that the chlorophyll *a* and chlorophyll *b* content were significantly enhanced by 90 % and 102 % respectively. Moreover, upon further analysis, it was also reported that the application of crude polysaccharides not only improved the chlorophyll content but also stabilized the membrane which eventually improved the plant's adaptability to altering temperatures (Rachidi et al., 2020). Many phenolic compounds are also produced by cyanobacteria. The phenolic compounds include gallic acid, protocatechuic acid, caffeic acid, and epicatechin (Safar et al., 2015). Moreover, flavonoids, terpenoids, steroids, saponins, and alkaloids are also produced by cyanobacteria (Chigayo et al., 2016). To encapsulate, cyanobacteria's role in improving the bioavailability of N and P, enrichment of photosynthetic pigments, phycobiliproteins, and secondary metabolites altogether helps in improving plant growth under extreme environmental conditions. Some of the species that produce different classes of metabolites are listed in Table 2.

### 3.4. The role of CBF on plant tolerance to biotic stresses

Cyanobacteria can produce different metabolites that are tested positive for their antimicrobial activities against pathogenic bacteria, and fungi. Similarly, it is also involved in antioxidant and anticancer activities (Kar et al., 2022). Almost 40 % of the known cyanobacterial secondary metabolites are reported to have both antimicrobial and anticancer activities (Nainangu et al., 2020). Cyanobacteria are widely used against plant pathogens and scientists are trying to discover novel metabolites and study their role as antimicrobial agents. To date, only some of the antimicrobial compounds have been structurally

**Table 2**

Classes of metabolites produced by cyanobacterial species and their effects on plants.

| Class           | Metabolites                                                                                                            | Function                                                                                                                                                                                                                                                     | Cyanobacterial species                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Reference                                                                                                               |
|-----------------|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Phytohormones   | IAA, siderophores, gibberellins, auxins, cytokinin                                                                     | Plant growth regulation and development (cell division, elongation), Tissue differentiation, Light, gravity, and pathogenic responses, Fe scavenger, Improve bioavailability of essential elements, Organ development and environmental response regulation, | <i>Nostoc entophyllum</i> ,<br><i>Synechocystis</i> sp.,<br><i>Anabaena variabilis</i> ATCC 29413,<br><i>Nostoc</i> sp.,<br><i>Microcystis aeruginosa</i> NIES-843,<br><i>Oscillatoria</i> ,<br><i>Chroococcidiopsis</i> sp.,<br><i>Phormidium</i> sp., <i>Trichormus</i> SAB-M304,<br><i>Nostoc</i> SAB-M251, <i>Calothrix ghosei</i> ,<br><i>Cylindrospermum</i> , <i>Westiellopsis prolifica</i> ,<br><i>Gloeotheca</i> ,<br><i>Nostoc</i> sp., <i>phaerospermopsis torques-reginae</i> ITP-024, <i>Aphanizomenon flos-aquae</i> , <i>Hapalosiphon intricatus</i> , <i>Microcystis aeruginosa</i> , <i>Cylindrospermum musciola</i> ,<br><i>Westiellopsis</i> sp. | (Ahmed and Holmström, 2014; Castro-Camba et al., 2022; Frébortová et al., 2017; Fu et al., 2015)                        |
| Peptides        | Polyamines/amino acids                                                                                                 | Plants defense, plant growth and developmental regulations                                                                                                                                                                                                   | <i>Nostoc</i> sp., <i>phaerospermopsis torques-reginae</i> ITP-024, <i>Aphanizomenon flos-aquae</i> , <i>Hapalosiphon intricatus</i> , <i>Microcystis aeruginosa</i> , <i>Cylindrospermum musciola</i> ,<br><i>Westiellopsis</i> sp.                                                                                                                                                                                                                                                                                                                                                                                                                                 | (Fidor et al., 2019; Gkelis et al., 2015; Nandagopal et al., 2021; Sanz et al., 2017)                                   |
| Polysaccharides | $\beta$ -glucans, lipopolysaccharides, carrageenans, chitin                                                            | Defense mechanisms, plant growth and regulation, conditioning agent,                                                                                                                                                                                         | <i>Phormidium</i> sp., <i>Anabaenopsis. circularis</i> ,<br><i>Oscillatoria</i> sp., <i>N. muscorum</i> , <i>A. nidulans</i> ,<br><i>A. platensis</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | (Bouyahya et al., 2024; Rossi and De Philippis, 2015; Shamshina et al., 2020; Singh, 2014)                              |
| Phenolics       | Gallic acid, protocatechuic acid, caffeic acid, epicatechin, flavonoids, terpenoids, steroids, saponins, and alkaloids | Antioxidant, anti-lipid peroxidative, anti-microbial, lignin synthesis, cell expansion regulation, maintains osmotic balance, defense mechanisms, signaling molecules                                                                                        | <i>Neochloris oleobundans</i> , <i>Phormidium</i> sp.,<br><i>Wilmottia murravi</i> , <i>Anabaena</i> sp., <i>Calothrix</i> ,<br><i>Oscillatoria</i> , <i>Nostoc</i> sp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | (Almendinger et al., 2021; Kollmen and Strieth, 2022; Mehmood et al., 2021; Singh and Sharma, 2015; Zhang et al., 2022) |

characterized, allowing room for further experimentation. Demay et al. reported that approximately 1630 molecules are extracted from cyanobacteria and are grouped into 260 different families of metabolites. Around 46.5 % (153 families) of the metabolite families are produced by the order Oscillatoriales, 29.7 % (98 families) by Nostocales, 10.3 % (34 families) by Chroococcales, and 9.4 % (31 families) by Synechococcales (Demay et al., 2019). The antibacterial activities of cyanobacteria are attributed to the presence of various compounds belonging to different classes. Most of the antibacterial compounds known so far belong to the class of alkaloids, fatty acids, pigments, terpenoids, phenolics, aromatics, cyclophanes, paracyclophanes, macrolides, indole, peptide, and polyphenyl ethers (Rojas et al., 2020). These chemical compounds work by inhibiting the growth and development of pathogenic microbes (Rojas et al., 2020).

Some examples of the biocidal activity of cyanobacteria against fungi, bacteria, oomycetes, and nematodes are shown in Table 3. Cyanobacteria prevent plant infections due to the biological agents which in return affect plant health, overall growth, and yield. Hence, cyanobacteria can be used as a biocontrol agent to protect plants from phytopathogenic diseases and to improve their health (Ramakrishnan et al., 2023; Sithole et al., 2023). It is imperative to say that biocontrol agents replace chemical-based pesticides that are source of contaminating water, soil, and vegetation (Aktar et al., 2009).

### 3.5. The role of CBF on soil properties and fertility

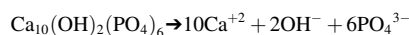
Cyanobacteria inoculation improves the bioavailability of N which is an essential element for the growth and development of plants. The predominant origin of N found in the soil arises from the biological fixation processes carried out by microorganisms (Mahanty et al., 2017). These nitrogen fixing cyanobacteria, if applied in the farms regularly (3–4 crop seasons), occupy the field soil permanently (Esteves-Ferreira et al., 2017). A study reported that the application of N<sub>2</sub>-fixing cyanobacteria in field enhanced the crop yield by improving the photosynthetic rate, stomatal conduction, and CO<sub>2</sub> concentration within the cells (Grzesik et al., 2017). *Oscillatoria angustissima*, and *Nostoc entophyllum*, two N<sub>2</sub>-fixing cyanobacteria improved the yield of *Pisum sativum* L and decreased the requirement of chemical fertilizer by 50 % (Osman et al., 2010). Moreover, *Phormidium ambiguum* application on sandy soil improved nitrate concentration in the soil by 15 % (Chamizo et al.,

2018).

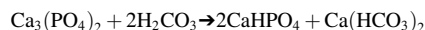
Cyanobacteria enhance the bioavailability of P to the host plants with the help of its insoluble organic phosphate solubilizing and mobilizing capability through phosphatase enzymes (Hong et al., 2021). P is a fundamental soil element necessary for the growth and development of plants and other species and is available in soil in both organic and inorganic forms (Ibrahim et al., 2022). But plants struggle to absorb it due to the complex structures it forms with other molecules such as Ca, Fe, or aluminum (Al) (Vaishampayan et al., 2001). In both scenarios, cyanobacteria play a crucial role in improving the accessibility of N and P for absorption. Cyanobacteria enhances P through the solubilizing immobile inorganic phosphate.

The soil P content is very small; 0.05 % of which only 0.1 % is bioavailable (Zhu et al., 2011). The P is made bioavailable from different insoluble forms such as iron (III) phosphate (FePO<sub>4</sub>), aluminum phosphate (AlPO<sub>4</sub>), hydroxyapatite [Ca<sub>5</sub>(PO<sub>4</sub>)<sub>3</sub>OH], and calcium phosphate [Ca<sub>3</sub>(PO<sub>4</sub>)<sub>2</sub>] (Zhu et al., 2018). Three well-known mechanisms are described below through which solubilization of phosphates is achieved.

The first mechanism is achieved by the production of Ca<sup>2+</sup> ions which act as potential chelators and it leads to the solubilization of phosphate compounds (Rai et al., 2019). The reaction of the mechanism is indicated in the following equation:



The second mechanism is the production and release of organic acids such as H<sub>2</sub>CO<sub>3</sub>. The organic acids begin to solubilize the phosphate compounds, thus solubilizing the phosphate molecules. pH plays a role in solubilizing the phosphate compounds as the organic acid production lowers the pH of the microenvironment (Kalayu, 2019). The reaction of the mechanism is given below:



The third mechanism does not make the phosphate readily available during the lifetime of cyanobacteria. In this process, organic phosphate is solubilized by cyanobacteria, and the resulting phosphate is utilized by it for its nutritional purposes. Once the cyanobacteria are dead, the phosphate is released into the soil and is bioavailable for the plants and other microbes to be taken up (Kumar Gupta et al., 2016).

Cyanobacteria regulate the biogeochemical cycles as it is directly involved in the fixation of CO<sub>2</sub> thus, involved in the cycling of C in soil.



**Table 3**

Different studies reporting the biocidal activities of extracts of cyanobacteria species on fungi, oomycete, bacteria, and nematodes.

| Cyanobacteria species/family/genus                                                                                                                   | Extract type                             | Species inhibited                                                                                                                                                                                                                                                           | Reference                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Genus   <i>Oscillatoria</i> , <i>Anabaena</i> , <i>Nostoc</i> , <i>Nodularia</i> , and <i>Calothrix</i>                                              | Petroleum ether and methanolic           | <i>Phytophthora capsici</i>                                                                                                                                                                                                                                                 | (Kim, 2006)                                    |
| Family   <i>Oscillatoriaceae</i> and <i>Scytonemataceae</i>                                                                                          | Organic                                  | <i>Fusarium fujikori</i> , <i>Rhizoctonia oryzae</i> , and <i>Curvularia oryzae</i>                                                                                                                                                                                         | (Gvns et al., 2011)                            |
| <i>Oscillatoria princeps</i>                                                                                                                         | Diethyl ether                            | <i>Pseudomonas aeruginosa</i> and <i>Aspergillus flavus</i>                                                                                                                                                                                                                 | (Marrez and Sultan, 2016; Marrez et al., 2022) |
| <i>Nostoc</i> ATCC 53789                                                                                                                             | Methanolic                               | <i>Penicillium expansum</i> , <i>Rosellinia</i> sp., <i>Armillaria</i> sp., <i>Sclerotinia sclerotiorum</i> , <i>Phytophthora cambivora</i> , <i>P. Cinnamomi</i> and <i>Caenorhabditis elegans</i>                                                                         | (Biondi et al., 2004)                          |
| <i>Trichormus variabilis</i> , <i>Tolypothrix tenuis</i> , and <i>Nostoc linckia</i>                                                                 | –                                        | <i>Sclerotinia</i> , <i>Pythium</i> , <i>Phaeoramularia</i> , <i>Phytophthora</i> , <i>Alternaria alternata</i> , <i>Fusarium graminearum</i> , <i>Botrytis cinerea</i> , and <i>Rhizoctonia P. ultimum</i> , <i>Botryodiplodia theobromae</i> , and <i>Fusarium solani</i> | (Horváth et al., 2023)                         |
| <i>Anabaena</i> sp., <i>Dunaliella</i> sp., and <i>Oscillatoria nigroviridis</i>                                                                     | Methanolic                               | <i>Alternaria alternata</i> , <i>Aspergillus niger</i> , <i>Botrytis cinerea</i> , and <i>Fusarium solani</i>                                                                                                                                                               | (Senousy et al., 2022)                         |
| <i>O. acutissima</i> , <i>Spirulina major</i> , <i>O. planctonica</i> , <i>O. brevis</i> , <i>Anabaena variabilis</i> , and <i>Nostoc verrucosum</i> | Hydroalcoholic                           |                                                                                                                                                                                                                                                                             | (Turcan and Syrbu, 2022)                       |
| <i>Spirulina</i>                                                                                                                                     | Aqueous                                  | <i>Cladosporium</i> sp., <i>F. culmorum</i> , and <i>A. solani</i>                                                                                                                                                                                                          | (Bencheikh et al., 2022)                       |
| <i>Tolypothrix</i> sp., <i>Nostoc</i> sp., and <i>Pseudanabaena</i> sp                                                                               | Methanol: Acetone: Diethyl ether (5:2:1) | <i>Ralstonia solanacearum</i> , <i>Fusarium oxysporum</i> , and <i>Sclerotium rolfsii</i>                                                                                                                                                                                   | (Bagul et al., 2022)                           |
| <i>Scenedesmus obliquus</i> and <i>Phaeodactylum tricornutum</i>                                                                                     | Aqueous                                  | <i>Sclerotium rolfsii</i> , <i>Rhizoctonia solani</i> , <i>B. cinerea</i> , and <i>A. alternata</i>                                                                                                                                                                         | (Schmid et al., 2022)                          |
| <i>Anabaena minutissima</i>                                                                                                                          | Aqueous                                  | <i>B. cinerea</i>                                                                                                                                                                                                                                                           | (Righini et al., 2021)                         |
| <i>Westiellopsis</i> sp.                                                                                                                             | Methanolic                               | <i>F. oxysporum</i> and <i>Sarocladium oryzae</i>                                                                                                                                                                                                                           | (Kiruthiga and Karthiba, 2020)                 |
| <i>N. linckia</i> , <i>A. variabilis</i> , and <i>Spirulina</i> sp.                                                                                  | Methanolic                               | <i>S. sclerotiorum</i>                                                                                                                                                                                                                                                      | (El-Khateeb and Nehela, 2019)                  |
| <i>Anabaena</i> spp., <i>Tolypothrix</i> spp., <i>Nostoc</i> , and <i>Trichormus</i>                                                                 | Aqueous                                  | <i>Pythium ultimum</i> and <i>Clavibacter michiganensis</i> subsp. <i>Michiganensis</i>                                                                                                                                                                                     | (Toribio et al., 2021)                         |
| <i>Anabaena</i> spp. and <i>A. laxa</i>                                                                                                              | –                                        | <i>Pythium aphanidermatum</i>                                                                                                                                                                                                                                               | (Gupta et al., 2010; Prasanna et al., 2008)    |
| <i>A. oscillarioides</i>                                                                                                                             | Aqueous                                  | <i>Pythium debaryanum</i>                                                                                                                                                                                                                                                   | (Chaudhary et al., 2012)                       |
| <i>Synechococcus elongatus</i>                                                                                                                       | Aqueous                                  | <i>Pantoea vagans</i>                                                                                                                                                                                                                                                       | (El Semary et al., 2022)                       |

**Table 3 (continued)**

| Cyanobacteria species/family/genus                                                                                                                                                                                                                                                         | Extract type           | Species inhibited                                                                                                                                                                                                                                                                                                            | Reference               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <i>Spirulina platensis</i>                                                                                                                                                                                                                                                                 | Methanolic             | <i>Staphylococcus aureus</i> , <i>Streptococcus epidermis</i> , <i>Streptococcus pyogenes</i> , <i>Bacillus cereus</i> , <i>Proteus mirabilis</i> , <i>Escherichia coli</i> , <i>Pseudomonas aeruginosa</i> , <i>Vibrio cholerae</i> , <i>Salmonella typhi</i> , <i>Klebsiella pneumoniae</i> , and <i>Shigella flexneri</i> | (Usharani et al., 2015) |
| <i>Oscillatoria</i> sp.                                                                                                                                                                                                                                                                    | Methanolic and Aqueous | <i>Meloidogyne incognita</i>                                                                                                                                                                                                                                                                                                 | (Ghareeb et al., 2022)  |
| <i>Trichormus variabilis</i> and <i>Nostoc punctiforme</i>                                                                                                                                                                                                                                 | Aqueous                | <i>Meloidogyne hapla</i>                                                                                                                                                                                                                                                                                                     | (Haque et al., 2022)    |
| <i>Synechococcus nidulans</i> , <i>Anabaena fertilissima</i> , <i>Aulosira pseudoramosa</i> , <i>Micochaete</i> sp., <i>Nostoc commune</i> , <i>Oscillatoria fremyi</i> , <i>Lyngbya</i> sp. <i>Phormidium molle</i> , <i>Tolypothrix phyllophila</i> , and <i>Westiellopsis prolifica</i> | Aqueous                | <i>M. incognita</i>                                                                                                                                                                                                                                                                                                          | (Sharma et al., 2007)   |
| <i>S. nidulans</i>                                                                                                                                                                                                                                                                         | –                      | <i>M. incognita</i> , <i>M. graminicola</i> , <i>Rotylenchulus reniformis</i> , <i>Heterodera avenae</i> , and <i>Heterodera cajani</i> .                                                                                                                                                                                    | (Holajjer et al., 2012) |

CO<sub>2</sub> sequestration by cyanobacteria has gained importance over the years as their CO<sub>2</sub> fixing rate is 10 to 50 times faster than terrestrial plants. Due to this reason, they are considered to be one of the top candidates that converts higher amount of solar energy and assimilates CO<sub>2</sub> through photosynthesis (Singh et al., 2016). Cyanobacteria inoculation in soil has been reported to improve the oxidizable and soluble C (de Mulé et al., 1999). Once dead, the biomass of cyanobacteria and its exopolysaccharides released are decayed thereby increasing the organic content of the soil along with the overall humus content (Román et al., 2018).

Potassium (K) is another important and abundant inorganic cation that is required by plants for its optimal growth. It ranks third among the major required macronutrients by plants (Lodi et al., 2021). K is a significant role player in activating enzymes that are important in protein synthesis and N and C metabolism in plants (Marschner, 2011). K is involved in stimulation and control of ATPase enzyme within the plasma membrane. Upon stimulation, it generates acid which triggers and initiates loosening of cell wall as well as activation of enzyme hydrolase thereby promoting the growth of plants' cell (Oosterhuis et al., 2014). Cyanobacterial biomass is composed of certain microelements that are essential for the growth of plants such as K, magnesium (Mg), sulfur (S), and Fe. Through the symbiotic association with plants, cyanobacteria assist plants in their metabolic processes (Gonçalves, 2021; Gonçalves et al., 2023). Similarly, soil inoculated with cyanobacteria have improved the percentage of Fe, Zn, Mn, and Cu not only in soil but also in plant parts (Manjunath et al., 2016; Prasanna et al., 2015b; Rana et al., 2012).

The diversity of soil microbiome is enhanced by the inoculation of

cyanobacteria as it improves the soil nutrients, stability, water content, and structure (Alvarez et al., 2024). With the improvement in all these factors, the colonization of other microbes and organisms is facilitated. For instance, in an arid region, the facilitation of lichens and mosses enhances the survival rate of annual and vascular plants. Similarly, the associated microfauna is also enriched due to the high organic C content of the soil. Well-developed and established soil biocrusts are usually enriched with cyanobacteria and are considered indicators of biodiversity, fertility, and soil health (Bowker et al., 2010).

### 3.6. Restoration of degraded ecosystem

#### 3.6.1. Cyanobacteria as pioneer species in land restoration

Ecosystem degradation is one of the most critical environmental problems the world is facing today. Approximately 20 billion ha of land in the world is degraded which covers about 22 % of the farmlands, forests, and prairie. A huge percentage of croplands, about 5 million ha, are not able to produce crops annually due to land degradation. If the number keeps on increasing at this pace, in the coming 20 years, one-third of the arable land will no longer be available for use (Ren et al., 2007). A degraded ecosystem is either formed naturally or by anthropogenic factors. In such degraded ecosystems, all the complex mechanisms and systems such as biodiversity and productivity are minimized in addition to the disturbed soil and microbial interactions. All the associated ecosystem services are no longer productive in a degraded ecosystem. Ecosystem degradation and agricultural practices go hand in hand. If no arable land is available, it is obvious that the net production of crops and vegetables, and other agricultural processes will also decline. Many efforts are underway to restore degraded ecosystems, of which the use of cyanobacteria is one (Hooda et al., 2023; Román et al., 2020). Recently, the role of cyanobacteria in land restoration has been studied. Because of the phenomenal characteristics of cyanobacteria, they are the first ones to colonize biocrusts. Hence, the colonization of cyanobacteria in biocrusts will enhance nutrient recycling thereby enhancing soil fertility. Cyanobacteria can withstand harsh environmental conditions that include, water shortage, extreme temperature, salinity, and pH (Chua et al., 2020). Moreover, the metabolites produced by cyanobacteria are very beneficial and can enhance the growth and development of plants by speeding up the seed germination and seedling growth stages. Since cyanobacteria is known to improve soil properties; water retention, porosity, and so on, it eventually improves the micro-environment for other soil communities and species like mosses and lichens (Chamizo et al., 2018). Furthermore, it helps maintain the survival conditions for plants living in arid regions.

The severe consequences of desertification i.e., partial, or complete disappearance of plants from an arid land is a kind of degraded land that happens due to different factors. Lan et al. (2014) investigated the impact of cyanobacteria on deserted land. They inspected the role of cyanobacteria in controlling desertification and restoring it by developing artificial crusts with 2 filamentous cyanobacteria: *Microcoleus vaginatus*, *Scytonema javanicum*, and a plantation site with *Salix*. After 3 years, in *Salix* planting with cyanobacteria inoculation, the crust was developing quickly and transformed into moss crust. The topsoil during that time was improving which also provided a platform for the regeneration of plants, specifically, vascular plants. Furthermore, 27 species of vascular plants were regenerated in the area inoculated with cyanobacteria. Other factors also played a role in regeneration such as shady regions showing more progress in vegetative cover improvement as compared to the non-shady regions (Lan et al., 2014). The succession of vegetation and replacement of dominant species starting from *Agriophyllum squarrosum* to *Leptocanna chimensis* indicates a successful reversal of desertification by inoculation of cyanobacteria. These findings show that the cyanobacterial crusts start forming rapidly after the introduction of cyanobacterial inoculation. Cyanobacterial crust formation binds the loose sand particles and builds cohesive soil surface that acts as a platform or a suitable microenvironment for vegetative

communities to grow.

#### 3.6.2. Enhancing soil aggregate stability

Cyanobacteria are reported to be used in improving and enhancing soil stability and fertility. Chamizo and his team tested cyanobacteria for stabilizing soil and improving its fertility (Chamizo et al., 2018). The team tested different textured soils: silt loam, sandy loam, loamy sand, and sandy soil. All of them had variations in particle size distribution and element composition (C, N). Cyanobacterium *P. ambiguum* wrapped around the soil particles by developing a network covering the surface of the soil just like a blanket, however, another cyanobacterium, *S. javanicum* produced a coral reef-like structure by bridging the soil particles. *P. ambiguum* produced relatively thinner filaments and the soil particles were connected to each other in specific patterns, however, *S. javanicum* formed thick filaments with irregular patterns. Additionally, the chlorophyll content of the soil before and after inoculation with cyanobacteria varied. As expected, the *P. ambiguum* chlorophyll content reached a maximum of 97 % in loamy sand as compared to 321 % in *S. javanicum* inoculated soil. Although inoculation of soil with cyanobacteria impacted the stability of the soil, each soil and inoculation type were impacted differently. The more stable soil was silt loam inoculated with *S. javanicum*. It was reported that increased soil stability was achieved with longer incubation time. For *P. ambiguum*, the soil stability was achieved to some extent, but to the very end of the incubation time. The inoculation of soil with cyanobacteria also enhanced its penetration resistance. Moreover, the EPS content in the soil also significantly increased followed by the C increase specifically in the sandy soil. Similarly, N was seen significantly increased in all soil types except sandy loam. The results of this study provided insights into the role of cyanobacteria in restoration of degraded land. *S. javanicum* and *P. ambiguum* improved soil fertility and stability. However, it is important to select species with outstanding traits to help restore degraded ecosystems (Chamizo et al., 2018).

In yet another research, a different technique was devised to carry out land restoration using cyanobacteria. Cyanobacteria in pellet form were used to inoculate soil. Two strains of N<sub>2</sub>-fixing cyanobacteria *Nostoc* and *Scytonema*, and one non N<sub>2</sub>-fixing cyanobacterium from genus *Leptolyngbya* were used. In this study, the team found that the pellets were successfully dissolved in all the treatments. *Scytonema* and all three strains together showed a significant increase in chlorophyll *a* content indicating the success of a novel technology of restoring degraded land by inoculating cyanobacteria pellets which would allow the contact time to be higher, thus, more survival time (Román et al., 2020).

In summary, cyanobacteria help in the restoration of degraded lands by developing the platform for other species to grow as first colonizers. Similarly, it can reverse the process of desertification through the formation of biological crust, soil stabilization, and improvement of other associated factors.

### 3.7. Enrichment of soil microbial communities

Arid regions or drylands are occupied by BSC. BSC growing in arid regions comprises mainly of cyanobacteria, algae, lichens, bryophytes, and heterotrophic bacteria (Chowaniec et al., 2023). The species diversity and composition in dryland areas are dependent on the total moisture content and the prevailing temperature. BSCs are well-studied for their roles in arid ecosystems provided that they fix approximately 30 % of the total biologically fixed global N (Elbert et al., 2012). Cyanobacteria are the pioneer microorganisms among the species that make the BSC due to their distinct features (Sandzewicz et al., 2023). Different types of cyanobacteria occupy the BSC including filamentous cyanobacteria, specific biocrust inhabitants, and cyanobacteria with different origins/sources (Büdel et al., 2016). The filamentous cyanobacteria through its filaments self-organize and build biofilms and bio mats, composed of multiple symbiotic communities. These biofilm/bio mats

with their symbiotic associations are found on a large scale in different ecological niches (Faluweki and Goehring, 2022). The second group of cyanobacteria prefers to live in biocrusts and hence are involved in the ecological uplifting by contributing to the C and N cycle (Hagemann et al., 2016). The third group, however, has a different origin but occurs in biocrusts, thus enhancing the ecological processes. After the colonization of cyanobacteria, other heterotrophic microbes start to grow due to improved environmental conditions which restrict the growth of cyanobacteria. These biocrusts further enhance the soil microbial diversity by improving bacteria-fungi and algae-bacteria interactions (Pombubpa et al., 2020; Ramanan et al., 2016).

### 3.8. The role of cyanobacteria in ecological succession and restoration of natural habitats

The co-occurrence of cyanobacterial associations not only enriches the microbiota of the soil but also improves the soil status. Since cyanobacteria is the first one to colonize and initiate the formation of BSC, 16S rRNA molecular-based analysis has proved the existence of cyanobacteria in the starting stage of succession, however, they disappear in the later phases of development. For instance, in desert soil of Dalateqi county, Inner Mongolia, where the organic C and total N are low, cyanobacteria *Microcoleus vaginatus* Gom. and *Scytonema javanicum* Born et Flah improved the soil fertility by 5 folds (Wang et al., 2009). Similarly, another study reported that *Anabaena doliolum* HH-209, *Cylindrospermum sphaerica* HH-202, and *Nostoc calcicola* HH-201 in semi-arid soil of Hisar, India improved and enhanced the C and N mineralization by directly influencing the soil microbial activities (Nisha et al., 2007).

Cyanobacteria are not only effective in improving soil fertility and chemistry but are also one of the most significant microbes that is used to remove HMs from polluted environments, hence, restoring the natural habitats. The overall process of remediation is environmentally friendly and user friendly, as well as energy saving as compared to other conventional technologies. Utilizing cyanobacteria for remediation purposes is an efficient and recognized biological technique due to its larger surface area and volume ratio, larger mucilage volume, higher binding ability with up to 10 % of their biomass, excessive binding sites, easy availability, uncomplex nutritious requirements, higher biomass production, adaptability to extreme environmental conditions, and ability to grow in a variety of habitats (Anjana et al., 2007; Igiri et al., 2018; Kaplan, 2013; Leong and Chang, 2020; Suresh Kumar et al., 2015). Different species of cyanobacteria are isolated from HM contaminated sites indicating their resistance to such environments. Different cyanobacterial species are used to remove HMs from contaminated sites using a variety of technologies. In one study, *Oscillatoria* sp. H1 isolated from Mogan Lake was used to remove cadmium (Cd) by the help of heat inactivated Ca-based immobilized alginate beads as support with maximum biosorption capacity of 27.5 mgg<sup>-1</sup> (Katurcioğlu et al., 2008). *Nostoc minutum* and *Anabaena spiroides* when mixed with fertilizers are reported to remove lead (Pb), cadmium (Cd), and nickel (Ni) from contaminated soil of Nubaria, Egypt (Al-Sherif et al., 2015). In another study, *Arthrospira platensis* immobilized in alginate, silica gel, and agarose remediated soil from Pb with maximum biosorption capacities of approximately 424.99, 272.66, and 2.68 mgg<sup>-1</sup> respectively. (Duda-Chodak et al., 2013). *Graesiella emersonii*, isolated from Ab-e-Siah hot spring, Ramsar, Iran was used as an efficient biosorbent against the isotopes of radium and uranium (<sup>226</sup>Ra and <sup>238</sup>U). The maximum absorption capacities reported for the strain was 69 kBqg<sup>-1</sup> for <sup>226</sup>Ra and 530 mgg<sup>-1</sup> for <sup>238</sup>Ua (Heidari et al., 2017). In another study, <sup>226</sup>Ra, <sup>238</sup>U, and Cd were biosorbed by the biomass of *Nostoc punctiforme* A.S/S4 and *Chroococcidio psis thermalis* S.M/S9. The maximum adsorption capacities of the adsorbed radionuclides (<sup>226</sup>Ra and <sup>238</sup>U) by *N. punctiforme* were 37 kBqg<sup>-1</sup> and 630 mgg<sup>-1</sup> while 55 kBqg<sup>-1</sup> and 730 mgg<sup>-1</sup> for *C. thermalis*. The maximum adsorption capacity for Cd were 160 mgg<sup>-1</sup> for *N. punctiforme* and 225 mgg<sup>-1</sup> for *C. thermalis* (Heidari et al., 2018). *Nostoc* sp. was isolated from coal mine water sample in Meghalaya,

India. The strain removed Cr with maximum adsorption capacity of 20 mgg<sup>-1</sup>. (Warjri and Syiem, 2018). In another research study, living cyanobacteria *Chroococcus multicoloratus* Wood and *Oscillatoria tri-choides* Szafer were used to remove Pb<sup>2+</sup> from an aqueous solution. The strains were isolated from polluted industrial sector in Mangalore, India. The maximum sorptive capacity of Pb<sup>2+</sup> removal by *C. multicoloratus* and *O. trichoides* were 178.57 mgg<sup>-1</sup> and 106.38 mgg<sup>-1</sup> (Miranda et al., 2013). The significant N<sub>2</sub> fixing cyanobacteria strains; *Scytonema* sp., *Nostoc muscorum*, *Cylindrospermum* sp., and *Anabaena* sp., are reported to remove 7.110 µgg<sup>-1</sup>, 3.88 µgg<sup>-1</sup>, 3.121 µgg<sup>-1</sup> and 2.999 µgg<sup>-1</sup> of 16 µM Al at pH 4 indicating their potential to be used as biofertilizers in acid rich soil (Shamim et al., 2022). Cyanobacteria are not always found to remove HMs but there are cases when the presence of HMs degrades the growth of cyanobacterial species. In the presence of HMs, cyanobacteria compete for the essential elements required for the enzymes and the transporters, hence it causes an imbalance in the functioning and growth of cells (Tottey et al., 2012). The toxicity of HMs depends on different factors including their concentration, oxidation state, and exposure time. For instance, a concentration of 0.2 mgL<sup>-1</sup> of Pb was toxic for the growth of *S. platensis* while for other species *Cyanothece* and *Anabaena*, 1 mgL<sup>-1</sup> affected the growth negatively (Choudhary et al., 2007; Heng et al., 2004). HMs contamination mainly affects the photosynthetic processes of cyanobacteria by attacking photosystems I and II (PS I & PS II) (Zhang et al., 2017). The activity of these PSs decreases due to the inhibition of their proteins such as the D1 protein of PS II (Zsiros et al., 2006). Furthermore, the respiratory process or rate is also decreased due to HM accumulation (Goswami et al., 2015). One of the most interesting features of cyanobacteria to resist HM toxicity is by the synthesis of metal-binding proteins also known as metallothionein's (MTs) (Kanwar et al., 2023). These proteins are composed of cysteinyl residues that bind metal ions to cysteine ligands by forming cysteinyl thiolate bridges. Generally, MTs are in association with Zn and copper (Cu) through which the cell homeostasis is maintained along with high protection against their toxicity, but MTs also have affinity towards other HMs that are toxic in nature like Cd and mercury (Hg) (Chatterjee et al., 2020).

In short, cyanobacteria in arid environments are the pioneers in colonizing the land and fostering the development of other microbial species contributing to the enrichment of microbial populations. Moreover, they also promote ecological succession as the arrival of different species brings along different types of associations and nutrient flux. Lastly, they remediate HM's polluted soil, thus, playing a role in restoration of natural habitats.

## 4. Response of cyanobacteria to climate change and the consequences in arid environment

Climate change mainly affects the precipitation rate, temperature, growing season lengths, and critical threshold events. Climate change, which impacts overall food production, poses an increased risk to food security (El-Beltagy and Madkour, 2012). With the ever-rising temperature, the crop yields decrease. Similarly, the minimum temperature during winter months also rises allowing pests to grow and survive for longer periods, hence affecting agriculture (Chen et al., 2020). The growth of microalgae such as cyanobacteria and the production of its secondary metabolites are affected by different environmental stressors. Under different stress conditions such as oxidative and salinity stress, the production of photosynthetic pigments like carotenoids is upregulated in cyanobacteria. Moreover, other compounds (phenolics, thamin) that regulate the antioxidative properties of cyanobacteria are enhanced under abiotic stress (Assunção et al., 2023). There exists a gap in the understanding of changing climatic conditions and cyanobacterial growth however, the effects of climatic changes are mimicked in studies on BSCs which are composed of communities of mosses, lichens, heterotrophs, and cyanobacteria. These stimulation studies might help in understanding the response of cyanobacteria to the changing climate. In one such study, it was mentioned that frequent but intermittent



precipitation caused the moss in BSC to die while enhancing the growth of cyanobacteria (Reed et al., 2012). In another study, it was reported that the moss and lichens within the BSC faced mortality due to C starvation that was achieved due to higher water levels than normal (Bowker and Belnap, 2008). With a high mortality rate of moss and lichens, cyanobacteria dominate in the BSC which indicates the shift of BSC towards the development of early successional biocrust. Although cyanobacteria growth is not much affected, the C and N fixation processes in successional biocrust are relatively low with high leaching rates (Barger et al., 2005; Housman et al., 2006). Contrary to the previous studies, the mimicked climatic variation in precipitation rates either had no significant effect on the BSC or only a negative effect during a specific season (Escobar et al., 2012; Johnson et al., 2012; Maestre et al., 2013). Like the seasonal differences, certain studies conclude that the effects of climate change on the BSC communities are influenced based on their geographic position (Ladrón de Guevara et al., 2014; Zedda et al., 2011). Considering the increasing temperature, it is not yet clear whether cyanobacteria will have long lasting beneficial impacts of warm temperatures (Wagner and Adrian, 2009) or a reversed reaction could be seen in the future. Moreover, it is worth mentioning that the effects of climatic changes on terrestrial cyanobacteria are not deeply studied and still require much attention as compared to water-based cyanobacteria. It is well noted in different studies that the ocean warming caused by climatic changes is contributing largely to the increased occurrence of harmful algal blooms (HAB). Increased temperatures not only increase the growth of HABs but also extend their growth beyond their geographic limitations hence, exposing the toxins to much extended areas (Umme Tamanna and Zetty Norhana Balia, 2022). Similarly, increased drought periods will lead to reduced flows that would ultimately improve the stability of water column due to its higher residence time as a result of which cyanobacterial HABs would be exposed to light for longer time enhancing their growth (Berg and Sutula, 2015). Another aspect associated with higher growth of cyanobacterial HABs is the rise in nutrients level (eutrophication) due to climate change. A study conducted controlled experiments at ambient and warming temperatures of 20 °C and 25 °C in both the presence and absence of excessive nutrients. The effect of the rising temperatures and nutrients were analyzed by analyzing the chlorophyll-a content. It was found that the chlorophyll-a content of cyanobacteria increased up to 18 % at 25 °C while an increase of 9 % was found in nutrient pulse (Lürding et al., 2018). To recapitulate, it can be said that the growth of cyanobacterial HABs improves with elevated temperatures and nutrients as evident from the results of the experiment.

## 5. Limitations of cyanobacteria applications

### 5.1. Toxin production, risks, and detection limitations

Cyanobacteria not only produce secondary metabolites that have beneficial roles but also produce toxic compounds or metabolites that are harmful to the living beings as well as the environment (Ricciardelli et al., 2023). Cyanotoxins are bioactive compounds ranging from peptides, and cyclic amine alkaloids to non-protein amino acids (Zhang and Whalen, 2019). Based on the literature, the most studied cyanotoxins belong to non-terrestrial ecosystems. There are many factors that contribute to the concentration of cyanotoxins being released, however, they are not exactly known. The regulatory factors specifically environmental factors play a significant role in the selection of the ratio of toxic versus nontoxic gene expression, or the nature of toxins to be either free or bound to protein bodies (Buratti et al., 2017).

The toxin exposure routes could be both direct and indirect. In the direct route, contamination of drinking and other water bodies occurs while in the indirect route either the food items contain cyanobacteria, or the food has bioaccumulated cyanotoxins (Abdallah et al., 2021). The contamination of water bodies with cyanotoxins starts an unchecked cycle where the adjacent soil and plant systems are contaminated as

well. Consequentially, leading to the spread of toxins from water to soil, and then to humans and animals. Moreover, the use of contaminated water for irrigation is considered the root cause of cyanotoxins in the soil. The water is absorbed in the soil and has a higher probability of leaching into the groundwater (Melaram et al., 2022). From groundwater, it can diffuse to the surrounding water bodies. Similarly, the cyanotoxins from the irrigated-toxin-rich water are absorbed by the plant roots. These toxins start to accumulate in the fruits and leaves of the plants posing a threat to the living organisms (Bouaïcha and Corbel, 2016). Cyanotoxins alter the microbial structure and associated functions (Li et al., 2020). Similarly, it affects different microbial processes such as nitrification. Moreover, biocrust is also reported to have a certain concentration of cyanotoxins (Chatziefthimiou et al., 2021).

Soil is not only polluted by irrigation with contaminated water but also by directly using biomass of toxic cyanobacteria as biofertilizers. Cyanotoxins in soil are not studied deeply because of challenges such as efficient extraction procedures and quantification of the cyanotoxins. The most challenging step in cyanotoxins extraction from soil is to select a suitable extraction solvent that would efficiently trap and extract cyanotoxins from the soil aggregates (Schmidt et al., 2014). Cyanotoxins have a relatively stable structure that is not easily degraded in the environment. In soil, cyanotoxins are removed through different processes such as UV degradation and microbial degradation (Bouaïcha and Corbel, 2016). The light-assisted UV degradation is very well studied specifically in microcystins; however, in soil, it is not exactly known (Bouaïcha and Corbel, 2016). Moreover, it is proposed that in soil, cyanotoxin is adsorbed on the particles of soil. The pH of the soil also affects the binding affinity of the cyanotoxin with soil as it can alter not only the surface charges of the cyanotoxins but also the net charge of soil surface particles (Zhang et al., 2020). Research shows that the chemical property of soil also plays a role in removing toxins. For instance, soil with high concentrations of organic C (2.9 %) and clay content (16.1 %) is found to be more efficient in removing cyanotoxins as compared to other soil types like sandy soil (Miller and Fallowfield, 2001), thus a challenge for arid regions with higher percentage of sandy soil. Similarly, another research supported the same concept where the removal of microcystins from sandy soil with 98 % sand content was not possible. Furthermore, the research verified the importance of clay in the removal of toxins from the soil. The adsorption of microcystins and their degradation is achieved by the adsorption of toxins with the metallic ions present in the soil. In this way, if not removed, microcystins can stay for longer times (Chen et al., 2006). Additionally, it is reported that the soil microbial community also plays a significant role in the degradation of microcystins and other cyanotoxins. Among the total microcystins in the soil, a total of 14 % is only adsorbed on the soil surface, while the remaining is proposed to be degraded by the microbes present in the soil (Corbel et al., 2014). There exists a gap in the knowledge regarding cyanotoxins with a special reference to the availability of comprehensive and in-depth toxicological data on cyanotoxins.

Moreover, there are limitations in the efficiency of the techniques used to identify the toxins. There exists a variety of techniques and assays to determine the cyanotoxins that include cell count, different chromatography techniques (HPLC, LC-MS), antibody utilized techniques (ELISA), toxin gene determination (PCR, qPCR), and toxicity estimation-based techniques (cell and enzymatic) (Gaget et al., 2017). All the mentioned techniques have advantages but there are limitations too. In certain techniques, for instance, the cell count method, the accuracy and precision are relatively low. Similarly, they only indicate the existence of a potential cyanotoxin despite the assumption that this procedure can accurately determine the toxin production rate per cell. Some of the procedures (chromatography-based) are not cost-effective and might not detect all the congeners. The antibody-based techniques such as the ELISA and dip-stick, though considered highly reliable for detection of all relevant and existing congeners, are practically not the same. The assay used in this process does not provide a response that is proportional to the quantity of cyanotoxin produced. Moreover, the



matrix used in such techniques can potentially interfere with the determination of cyanotoxins. PCR-based techniques are reported to have low accuracy and interference in the determination of toxic genes. Similarly, a bigger limitation exists in the fact that toxigenic species with different gene sequences might not be detected using PCR or qPCR. Lastly, cell and enzymatic-based toxicity determinants also have lower precision and accuracy with potentially interfering factors (Gaget et al., 2017). Considering the advantages and disadvantages of each technique, it is important to carry out more research on the detection methods and to come up with a reliable and efficient approach. It is highly likely that one technique might work best for a group of cyanotoxins, but it might not prove to be efficient for another group.

## 5.2. Techno-economic feasibility

Using cyanobacteria on a large scale poses many technical challenges that mainly include the cultivation and harvesting of cyanobacteria. For large scale cultivation of cyanobacteria, three types of system are used: ponds, photobioreactors (PBR), and biofilm systems. In some cases, all the systems are used in a combination. Pond systems are either closed-pond or artificially made open-pond systems. Open-pond systems have low productivity as compared with closed-pond systems which are costly. Moreover, not all cyanobacterial species can be grown in an open-pond system under controlled growth conditions. The limitation of the open-pond system is low productivity or low biomass production due to fluctuating environmental conditions (Chisti, 2007; Pathak et al., 2018). Open-pond systems tend to be cost-efficient as they have low construction and maintenance costs as well as low energy consumption (Chabili et al., 2024). Closed-pond systems, on the other hand, are designed in a way that they work on capturing solar radiation, however, maintaining other factors such as oxygen level and temperature is critical. The biomass yield is generally high when a closed-pond system is used (Enzing et al., 2014). When compared with other systems, ponds are relatively cheaper, but they require a large land area. Cyanobacteria grown in ponds have poor CO<sub>2</sub> efficiency and they are more likely to be affected by weather conditions such as rain or evaporation. PBRs, on the other hand, are much favorable and are designed in a way to have large surface area to culture volume. The drawback of this system is to have heat loss when no light is present (Morita et al., 2001). The third type of system is biofilm, which is an inexpensive and unique technique to mass produce cyanobacteria. The type of system is selected based on the potential application (Aitken and Antizar-Ladislao, 2012). To prevent culture crash, techniques to monitor growth-associated factors are developed but are expensive (Khatoon and Pal, 2015; Pathak et al., 2018). Apart from the selection of an appropriate system for mass production of cyanobacteria, other components should also be considered to achieve a sustainable and cost-effective process. The two most important factors for growth of cyanobacteria at large scale are water and the nutrients required to grow biomass. For mass production of cyanobacteria, large volume of media is required which is not cost efficient, therefore, it is advisable and much more convenient to rely on the commercially available salt for preparing the media by mixing it with either tap or deionized water (Jepsen et al., 2019). Some reports suggest that the use of commercially available salt might not be preferable or suitable for growing microalgae due to lack of necessary nutrients, therefore, additional steps of adding micro and macro nutrients would be required. It is suggested that rather than using commercial salt with addition of nutrients, readily available sea water could be used that already has micro and macronutrients (Atkinson and Bingman, 1997). Interestingly, some researchers have reported the use of agricultural fertilizers to grow microalgae with net yield higher than or equal to the yield when traditional media is used (Brito et al., 2016; Karuppan et al., 2019; Martínez-Córdova et al., 2012). The use of agricultural fertilizers to grow microalgae can reduce the cost of media preparation by 95 % as there is no requirement of adding micro and macronutrients (Camacho-Rodríguez et al., 2020; Soares et al., 2018).

In the commercialization process of microalgae products, one of the major limitations is the cost required for large scale microalgal production. The cost for producing 1 kg dry weight ranges between € 290 (317.50 US \$) to € 570 (624.6 US \$) based on the production process used (Oostlander et al., 2020). This cost is mainly related to the mixing of culture volume, light, temperature, cleaning, and harvesting (Novoveská et al., 2023). There are additional charges associated apart from the biomass preparation, that includes the nutrient supply cost mainly N source. It is reported that for the annual production of 1.8 tons of biomass, it costs 16,160 € (17,720.25 US \$) for N source (Branco-Vieira et al., 2020). As an alternative, it is suggested to use wastewater to grow cyanobacteria to minimize the nutrients costs (Arashiro et al., 2018).

Cyanobacteria fix N<sub>2</sub> through its nitrogenase enzyme that help the plants to be able to grow in soil with less N hence, limiting the need to use chemical-based N fertilizers obtained at higher prices (Li et al., 2019). When used alone, without the addition of chemical fertilizers, cyanobacteria as biofertilizers tend to be inexpensive and cost one-third of the price used for chemical fertilizers (Prasanna et al., 2013). A study reports that the production cost related to the cultivation of rice decreased by half when 50 % of the recommended chemical nitrogen was substituted with cyanobacteria-based biofertilizer (Chittapun et al., 2018). Moreover, in another study it is reported that the use of cyanobacteria in different forms saves 50 % of chemical fertilizer particularly for wheat grain yield (Alvarez et al., 2024). Cyanobacteria biomass was prepared using different types of straw waste: *Azadirachta indica* (neem), *Nicotina tabacum* (tobacco), *Aegle marmelos* (bel), and *Oryza sativa* (rice) for use as biofertilizer. The biofertilizer was applied alone and in combination with chemical fertilizers. In both cases the grain and rice yield improved. Using cyanobacteria biofertilizer, 25 kg of Nha<sup>-1</sup> was saved (Jha and Prasad, 2006). In India, cyanobacteria have been used together with chemical inputs to cross check the cost of cultivation, net yield, and income benefits of farmers. Urea, Diammonium phosphate (DAP), potash, and zinc (kg ha<sup>-1</sup>) were used. It was found that the urea application when cyanobacteria was applied reduced by 25 % decreasing the cost of cultivation from 324 US \$ to 318 US \$ ha<sup>-1</sup> (Bhooshan et al., 2020). In China, study reports that the substitution of up to 30 % of nitrogen fertilizer with NFC could save up to 5.77 US \$ ha<sup>-1</sup> (Zhang et al., 2021).

It can be said that the features of cyanobacteria, especially N<sub>2</sub> fixing activity through nitrogenase enzyme, have made it one of the promising cost-effective candidates to be used as biofertilizers.

## 6. Concluding remarks and recommendations

The ability of cyanobacteria to grow in harsh climatic conditions is a positive indicator for its ability to be utilized as a sustainable approach towards sustainable agriculture. Cyanobacteria through its characteristic features enhance soil fertility and supports the growth of plants through improving the bioavailability of micro and macro-nutrients. Further, through its N<sub>2</sub> fixing ability, cyanobacteria can reduce the cost of expensive chemical N sources that are used as fertilizers despite their harmful impact on mother nature. This would benefit farmers from the poor and developing countries that cannot afford to manage costly synthetic fertilizers. The production of secondary metabolites and phytohormones by cyanobacteria assists in managing the growth regulations of plants along with activating defense mechanisms against various biotic and abiotic stressors. As a pioneer species, cyanobacteria can also be utilized for the restoration of degraded land.

Various studies on different aspects of cyanobacteria are listed in the literature, however, it is recommended that research should focus on climatic changes and their impacts on terrestrial cyanobacteria as much attention is given to the non-terrestrial species. We would also like to recommend that the techniques to detect cyanotoxins should also be emphasized so that new methods could be developed that are much more accurate, reliable, broad-range, and quick. Application of cyanobacteria at large scale globally to fulfill the food demands might be

hindered due to social and political barriers, therefore, these points should also be considered and justified by the authorized organizations.

### CRedit authorship contribution statement

**Shazia Bibi:** Writing – original draft, Methodology, Investigation, Conceptualization. **Imen Saadaoui:** Writing – review & editing, Supervision. **Amina Bibi:** Writing – original draft, Investigation, Conceptualization. **Mohammad Al-Ghouti:** Writing – review & editing, Supervision, Conceptualization. **Mohammed H. Abu Dieyeh:** Writing – review & editing, Supervision, Resources, Investigation, Conceptualization.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Data availability

No data was used for the research described in the article.

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